How many computers are in a Cat S22 Flip

About eleven, depending on how strict you are about what counts as one. The main chip (a Qualcomm QM215, a variant of the MSM8917) holds about seven separate processors, each running its own firmware; four more sit outside it.

Inside the main chip

#	Processor	Runs	Where it shows up in BENCtix-CAT
1	4 × Cortex-A53	Linux; TrustZone runs on the same cores in a hidden secure mode	"The phone"
2	Modem (Hexagon DSP)	Qualcomm's own operating system and the LTE stack	Calls, texts, IMS, and GPS: the location engine lives here (memshare, D078)
3	Audio and sensor DSP (Hexagon)	Audio processing and the sensor hub	The accelerometer, pressure and proximity sensors all go through it; it never slept until an audio clock was released
4	Pronto (a small ARM core)	The WiFi and Bluetooth firmware	WoWLAN; it woke the phone for Tailscale's packets (D082)
5	RPM (Cortex-M3)	Power management: votes on clocks and voltages for everyone else	The deep-sleep (vlow, vmin) counts in the battery log come from its memory
6	Venus	Video encoding and decoding firmware	The "venus" interrupt
7	GPU (Adreno 308)	Shaders, plus a small command processor running its own microcode	The browser's renderer and recovering from a GPU reset (D077)

Outside the main chip

#	Part	Why it counts
8	Touch controller (chipsemi, 0-002e)	A small microcontroller; the kernel log shows its "firmware 8020" at every wake
9	eMMC storage	Its own controller, usually an ARM core, manages the flash (wear levelling, bad blocks)
10	SD card	The same: a controller in the card
11	SIM card	A real smartcard computer: a CPU, memory, and usually a small Java operating system

Not counted

The PMIC (power chip) has the voltage-mode battery gauge and the charger logic, but it is fixed hardware, not a CPU. The same goes for the camera image pipeline, the crypto engine and the USB-C controller. The motion and light sensors are just sensors; the audio DSP does their thinking.

Who is in charge

Linux is only one of these, and not the one in charge. TrustZone, the RPM and the modem each run code Linux can't see or override; Linux mostly negotiates with them by passing messages. That is why so much of the port has been about message channels (SMD, QMI, QRTR) and firmware rather than drivers in the usual sense.